Sanjai V

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.21
Aditya Vidyashram	Class XII (Central Board Of Secondary Education)	2021	86.20%
The PSBB Millennium School	Class X (Central Board Of Secondary Education)	2019	81.40%

WORK EXPERIENCE

Indian Space Research Organisation(ISRO)

- Analyzed performance parameters and testing hardware for upgraded solid **rocket propulsion** systems.
- Modeled test setups using **SolidWorks** and conducted fluid flow simulations in **ANSYS Fluent**.
- Validated experimental data against **ANSYS Fluent** models, achieving **95%** computational accuracy.

Larsen & Toubro

- Developed a **vision-based navigation** algorithm suite for **real-time 3D** relative pose estimation.
- Simulated and optimized **multi-algorithm** tracking performance under diverse operational scenarios.
- Designed a closed-loop feedback controller in **MATLAB** and **Simulink** for precise motion control.

PROJECTS

Experimental Evaluation of Reinforcement Learning for Control of Underactuated Robotic Systems

- Investigating RL-based stabilization, beginning with a self-balancing robot in MuJoCo and on physical hardware.
- Planning to extend the control methodology to a laboratory helicopter UAV, evaluating performance against conventional controllers and under variations in system dynamics.

Vision-Based Navigation: Application to Spacecraft Docking and Quadcopter Aerial Manipulation

- Built a **MATLAB** vision pipeline (CLAHE, SIFT, SSD matching, RANSAC) to estimate **relative rotation and distance** for a **1P1R docking** mechanism. In the test case (90° rotation, 0.3 m), it returned 90.02° and 0.300 m.
- Sized the actuators, calculated torque requirements, and chose servo motors over steppers for finer angular resolution.
- Modeled the servo in **Simulink**, added a controller to control **angular displacement**, and validated the **docking motion** in **Simscape**.
- Extended the pipeline with **circularity-based shape recognition** and **grasp-angle selection** for a **quadcopter-mounted arm** for payload grasping.

CFD Analysis of Backward-Facing Step Flow

- Executed compressible flow simulations at **Mach 0.5 to 1.2** utilizing the **k- SST** turbulence model.
- Located reattachment from the **zero-crossing of wall shear stress**, about **7.9** for step height of H = 0.01 m and **5.5** for H = 0.03 m.
- Showed reattachment length changed under **9%** from Mach **0.5 to 1.2** but **roughly doubled** when the **step height tripled**.

Optimal Autonomous Control of an F1 Car with Active Aerodynamics

- Developed a high-fidelity **Nonlinear Model Predictive Controller** traversing the **Suzuka 130R** corner.
- Integrated 2nd-order wind-tunnel polars for **Selig S1223** to enable active downforce control.
- Achieved **stable path tracking** within a **±2.5 m lateral error** across a **300-step simulation** framework.

Thermal Mechanical Analysis of an Aircraft Wing

- Modeled a wing assembly (2 mm skin, 10 ribs, 4 I-beam spars) in **Fusion 360** for **thermo-structural FEA**.
- Executed **thermo-structural** simulations under **100 kPa** aerodynamic pressure and **40°C** thermal loading and compared stress.
- Found **Ti-6Al-4V** gave the lowest peak stress (71.5 MPa), **42%** below **Al-7075** (124.2 MPa) and **21%** below **CFRP** (90.3 MPa).

PUBLICATIONS

Circularity Parameter Utilized Vision-Based Navigation for Quadcopter's Robotic Arm

IEEE Internation conference on Emerging Technology in Autonomous Aerial Vehicles (ETAAV) - Aug. 2025

Coordinated Aerial Grasping Using Vision-Based Navigation for Aerial Manipulation

IEEE SPace, Aerospace and defenCE Conference - June 2026

SKILLS

CAD and Manufacturing: Fusion 360, SolidWorks, AutoCAD, 3D Printing.

FEA and CFD: Ansys (Fluent), Abaqus, Ansys (Workbench), Ansys (Mechanical).

Programming: Python (NumPy, SciPy, pytest), MATLAB, C, Arduino.

Simulation: MATLAB/Simulink, Simscape, STK, XFLR5, Open Rocket.

Software: LaTeX, Microsoft Office.

RELEVANT COURSES

Mechanics of Materials, Aerospace Structures 1, Aerospace Structures 2, **Finite Element Methods for Aerospace**, Dynamics and Vibration, **Air Breathing Engines**, Aerospace Propulsion, Fundamentals of Heat Transfer, Introduction to Thermodynamics, Compressible Fluid Flow, **CFD for Aerospace**, **Advanced CFD**, Machine Learning and Applications.

POSITIONS OF RESPONSIBILITY

Class Representative, Batch 2025 - 2027 M.Tech Aerospace Engineering, **fostering** communication for **15** students.

Served as **Teaching Assistant** for **75+ students** in **Control Systems** and **Flight Dynamics**, overseeing grading and exam invigilation.

Sports House Secretary, Organized **inter department** events during the 2023 - 2024 academic year.

Coordinated venue logistics, workshop operations, and participant management as **Department Volunteer** for the **Anokha** cultural fest.

SCHOLASTIC ACHIEVEMENTS

Awarded a **50% academic scholarship** during B.Tech for consistent academic excellence.

CERTIFICATIONS

Systems Tool Kit (STK) Certification | SS Technologies (4-Month Training)

EXTRACURRICULAR

1st Place in both the **Institute Football League** and **Milan (Inter-Hostel Championship)** in Football at IIT Hyderabad.

3rd Place in the **Formula 1 Quiz** conducted during Mechanical & Aerospace Engineering (MAE) Department Day at IIT Hyderabad.

LANGUAGES

Tamil (Native), **English** (Professional), **Hindi** (Elementary)